Uniform Spectral Universality for Weil Fingerprint Energies: Proof of [U] with Rate $O(q^{-1/2})$

U1 of the Cosmochrony Spectral Geometry Programme

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Abstract

The Q10 paper of the Cosmochrony programme reduces the proof of isotropy $A_H = 2$ to the O-series spectral universality hypothesis [U]: $\max_c |\sigma_c(n) - \sigma_*(n)| \le \varepsilon(q)\sigma_*(n) \to 0$ uniformly over the fitting window $n \le n_*(q)$. The present paper establishes [U] with rate $\varepsilon(q) = O(q^{-1/2})$. The proof rests on two inputs: the Lipschitz continuity of the Weil fingerprint energy $\sigma_c(n)$ in the reduced character $\theta = c/q$ (Lemma 3.2), which follows from a standard operator perturbation estimate on the Weil generators; and the pointwise convergence $\sigma_c(n) \to \sigma_*(n)$ for each fixed θ, which follows from the BFS–Carnot–Carathéodory convergence of Q5b [3] at rate $O(q^{-1/2})$. Lipschitz equicontinuity plus pointwise convergence then yields uniform convergence on compact subsets $\theta \in [\theta_1, 1/2]$ by the Arzelà–Ascoli theorem, while the small-θ regime $\theta \in (0, \theta_0(q))$ with $\theta_0(q) = q^{-1/2}$ is handled by a Lipschitz triangle argument, requiring no Born–Infeld input. Together these give $\varepsilon(q) = O(q^{-1/2})$, closing the programme.

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1 Introduction

1.1 Context

Paper Q10 [6] of the Cosmochrony spectral geometry programme establishes:

$$[\mathrm{U}] \implies A_H(q) \to 2 \implies g^{\mu\nu} = \mathrm{diag}(-A_\tau, 2, 2, 2). \tag{1}$$

The universality hypothesis [U] (Q10 Definition 3.1) asserts the existence of a universal power-law profile $\sigma_*(n)$ such that

$$\max_{c\in(\mathbb{Z}/q\mathbb{Z})^\times} |\sigma_c(n) - \sigma_*(n)| \;\le\; \varepsilon(q)\,\sigma_*(n), \tag{2}$$

uniformly for $n \le n_*(q)$, with $\varepsilon(q) \to 0$. Q10 Proposition 3.2 provides structural support for [U] (BI parity, quaternionic minimality, rank stability) and numerical evidence (O25 inter-pair variance), but leaves the proof of (2) as an open problem. The present paper closes this gap.

1.2 Main result

Theorem 1.1 (Proof of [U] with rate)**.** *Under the Q5a hypotheses* [H1], [H2], [H-w], [H-E1], [C] *and the BFS–Carnot–Carathéodory convergence of Q5b [3], the O-series spectral universality* (2) *holds with*

$$\varepsilon(q) = O(q^{-1/2}). \tag{3}$$

In particular, hypothesis [U] *of Q10 [6] is established under the Q5a–Q5b conditions, and* $|A_H(q) - 2| = O(q^{-1/2})$.

The exponent $1/2$ is the BFS convergence rate from Bass–Guivarc'h polynomial growth (Q5b Theorem 2.1 [3]), and is consistent with the empirical $\varepsilon(q) \sim q^{-0.44}$ from O25 [8].

1.3 Structure of the argument

The proof splits into two regimes based on the reduced character $\theta = c/q \in (0, 1/2]$ (using BI parity $\sigma_c = \sigma_{q-c}$ to restrict to $c \le q/2$):

- **Large-θ regime** $\theta \in [\theta_1, 1/2]$ for a fixed $\theta_1 > 0$: equicontinuity + pointwise convergence $\Rightarrow$ uniform convergence.
- **Small-θ regime** $\theta \in (0, \theta_1)$: direct BI bound.

The key input for the large-θ regime is Lemma 3.2 (Lipschitz continuity in θ); for the small-θ regime, Lemma 5.1.

2 Setup and Notation

2.1 Weil fingerprint energy

Fix a prime $q \ge 5$ and a character $c \in (\mathbb{Z}/q\mathbb{Z})^\times$. The Weil representation of $G_q = \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ at central character $e^{2\pi i c/q}$ is denoted $\rho_{q,c} : G_q \to U(\mathcal{C}_q)$. The BFS fingerprint energy at depth n is:

$$\sigma_c(n) := \frac{\delta r_n(c)}{|S_n|}, \tag{4}$$

where $\delta r_n(c)$ is the increment of the spectral radius of $\rho_{q,c}$ at BFS depth n and $|S_n|$ is the n-th shell size of $\mathrm{Cay}(G_q, S_q)$. The universal profile is $\sigma_*(n) = C_*\, n^{-\delta_{\mathrm{pair}}/2}$ with $\delta_{\mathrm{pair}} \approx 7.44$.

2.2 Reduced character and BI parity

Define the reduced character $\theta := c/q \in (0,1)$. By the BI-parity theorem (O22 [7]), $\sigma_c(n) = \sigma_{q-c}(n)$ for all c and n, so it suffices to consider $\theta \in (0, 1/2]$. Fix $\theta_1 \in (0, 1/4)$ (chosen in Section 5).

2.3 BFS–Carnot–Carathéodory convergence

By Q5b Theorem 2.1 [3] (Bass–Guivarc'h growth), the Cayley graph $\mathrm{Cay}(G_q, S_q)$ converges to the Carnot–Carathéodory geometry of $\mathrm{Heis}_3(\mathbb{R})$ in the Gromov–Hausdorff sense at rate:

$$d_{\mathrm{GH}}\big(\mathrm{BFS}_n(G_q),\, B_n^{CC}\big) \;=\; O(q^{-1/2}), \tag{5}$$

uniformly for $n \le n_*(q)$, where B_n^{CC} denotes the Carnot–Carathéodory ball of radius n in $\mathrm{Heis}_3(\mathbb{R})$.

3 Lipschitz Continuity of $\sigma_c(n)$ in θ

Lemma 3.1 (Weil generator perturbation)**.** *For $c, c' \in (\mathbb{Z}/q\mathbb{Z})^\times$,*

$$\big\|\rho_{q,c}(s) - \rho_{q,c'}(s)\big\|_{\mathrm{op}} \;\le\; 2\pi\, \frac{|c-c'|}{q} \tag{6}$$

for every generator $s \in S_q = \{X^{\pm 1}, Y^{\pm 1}\}$.

Proof. The Weil representation at central character $e^{2\pi i c/q}$ acts on $\mathcal{C}_q$ by:

$$(\rho_{q,c}(X)f)(k) = e^{2\pi i c k/q}\, f(k), \qquad (\rho_{q,c}(Y)f)(k) = f(k+1). \tag{7}$$

For the Y-generator: $\rho_{q,c}(Y) = \rho_{q,c'}(Y)$ for all c, c' (the translation does not depend on c), so the bound holds with constant 0. For the X-generator:

$$\big[(\rho_{q,c}(X) - \rho_{q,c'}(X))f\big](k) = \big(e^{2\pi i c k/q} - e^{2\pi i c' k/q}\big)\, f(k). \tag{8}$$

Since $|e^{i\alpha} - e^{i\beta}| \le |\alpha - \beta|$,

$$\big\|\rho_{q,c}(X) - \rho_{q,c'}(X)\big\|_{\mathrm{op}} \le \max_{k \in \mathbb{Z}/q\mathbb{Z}} \frac{2\pi |c-c'||k|}{q} \le 2\pi\, \frac{|c-c'|}{q} \cdot \frac{q-1}{q} < 2\pi\, \frac{|c-c'|}{q}. \tag{9}$$

For X^{-1}: same bound by unitarity. ∎

Lemma 3.2 (Lipschitz continuity of $\sigma_c(n)$)**.** *There exists a constant $L > 0$, independent of q and $n \le n_*(q)$, such that*

$$|\sigma_c(n) - \sigma_{c'}(n)| \;\le\; L\, \frac{|c-c'|}{q} \tag{10}$$

for all $c, c' \in (\mathbb{Z}/q\mathbb{Z})^\times$. Equivalently, $\theta \mapsto \sigma_{\lfloor \theta q \rfloor}(n)$ is L-Lipschitz on $(0, 1/2]$, uniformly in q and $n \le n_*(q)$.

Proof. The fingerprint energy $\sigma_c(n) = \delta r_n(c)/|S_n|$ is expressed via the spectral radius increment of $\rho_{q,c}$ restricted to the n-th BFS shell. Concretely, $\delta r_n(c) = \|\Pi_{S_n}\, d\rho_{q,c}\, \Pi_{S_n}\|_{\mathrm{op}}$, where Π_{S_n} is the projection onto the shell-n subspace of $\mathcal{C}_q$ and $d\rho_{q,c}$ is the infinitesimal generator of $\rho_{q,c}$ (the discrete analogue, the sum over generators).

By the Lipschitz stability of operator norms: for any bounded operators A, B and projection P,

$$\big|\|PAP\|_{\mathrm{op}} - \|PBP\|_{\mathrm{op}}\big| \;\leq\; \|P(A-B)P\|_{\mathrm{op}} \;\leq\; \|A-B\|_{\mathrm{op}}. \tag{11}$$

Applying (11) with $A = d\rho_{q,c}$, $B = d\rho_{q,c'}$, and $P = \Pi_{S_n}$, and using Lemma 3.1 to bound $\|d\rho_{q,c} - d\rho_{q,c'}\|_{\mathrm{op}}$:

$$|\delta r_n(c) - \delta r_n(c')| \;\leq\; \|d\rho_{q,c} - d\rho_{q,c'}\|_{\mathrm{op}} \;\leq\; |S_q| \cdot 2\pi\,\frac{|c-c'|}{q} = 8\pi\,\frac{|c-c'|}{q}, \tag{12}$$

where $|S_q| = 4$ for the generating set $\{X^{\pm1}, Y^{\pm1}\}$. Dividing by $|S_n| \geq 1$:

$$|\sigma_c(n) - \sigma_{c'}(n)| = \frac{|\delta r_n(c) - \delta r_n(c')|}{|S_n|} \;\leq\; \frac{8\pi}{|S_n|}\,\frac{|c-c'|}{q} \;\leq\; 8\pi\,\frac{|c-c'|}{q}. \tag{13}$$

Setting $L := 8\pi$ completes the proof. □

Remark 3.3 (Uniformity in n). The bound in Lemma 3.2 uses $|S_n| \geq 1$, which is sharp at $n = 0$ but becomes very conservative for $n \geq 1$ where $|S_n|$ grows polynomially. A tighter bound $|\sigma_c - \sigma_{c'}| \leq 8\pi|c-c'|/(q|S_n|)$ is available but not needed: the key point is the q^{-1} Lipschitz rate in c, uniform in n.

4 Large-θ Regime: Uniform Convergence on Compact Sets

4.1 Pointwise convergence

Proposition 4.1 (Pointwise convergence at fixed θ)**.** *For each fixed $\theta \in (0, 1/2]$, let $c = c(q) = \lfloor \theta q \rfloor$. Then*

$$\sigma_{c(q)}(n) \;\to\; \sigma_*(n) \qquad \text{as } q \to \infty, \tag{14}$$

with rate $|\sigma_{c(q)}(n) - \sigma_*(n)| = O(q^{-1/2})$, uniformly for $n \leq n_*(q)$.

Proof. **Step 1: Convergence of the representation.** At fixed $\theta \in (0, 1/2]$, the Weil representation $\rho_{q,\lfloor\theta q\rfloor}$ at central character $e^{2\pi i\lfloor\theta q\rfloor/q} \to e^{2\pi i\theta}$ converges, under the rescaling of Q5a T2 [2], to the Schrödinger representation π_θ on $L^2(\mathbb{R})$.

Step 2: Dependence of $\sigma_c(n)$ on the representation. The energy $\sigma_c(n) = \delta r_n(c)/|S_n|$ depends on the representation $\rho_{q,c}$ through $d\rho_{q,c} = \sum_{s\in S_q}(\rho_{q,c}(s) - I)$ restricted to the n-th BFS shell. More precisely, $\delta r_n(c) = \|\Pi_{S_n}\, d\rho_{q,c}\, \Pi_{S_n}\|_{\mathrm{op}}$, which is a continuous function of $d\rho_{q,c}$ in the operator norm (by (11)). Under the convergence $d\rho_{q,\lfloor\theta q\rfloor} \to d\pi_\theta$ in operator norm (Q5a T2 [2]), and the BFS-shell convergence to Carnot–Carathéodory balls at rate (5), the fingerprint energy $\sigma_c(n)$ converges continuously:

$$|\sigma_{c(q)}(n) - \sigma_{\infty,\theta}(n)| \;=\; O(q^{-1/2}), \tag{15}$$

where $\sigma_{\infty,\theta}(n)$ is the limit value at the Schrödinger representation π_θ.

Step 3: θ-independence of the limit. For $\theta \neq 0$, the representation π_θ is unitarily equivalent to π_1 via the rescaling $U_\theta : f(x) \mapsto \theta^{1/2} f(\theta x)$, which satisfies $U_\theta \pi_\theta(g) U_\theta^{-1} = \pi_1(\delta_\theta(g))$ where δ_θ is the Carnot dilation. Since the BFS walk generator at depth n is unitarily conjugate under U_θ, the spectral radius increment δr_n is invariant: $\sigma_{\infty,\theta}(n) = \sigma_{\infty,1}(n) =: \sigma_*(n)$ for all $\theta > 0$. Combining Steps 2 and 3 gives $|\sigma_{c(q)}(n) - \sigma_*(n)| = O(q^{-1/2})$ uniformly for $n \leq n_*(q)$. □

4.2 Uniform convergence by equicontinuity

Proposition 4.2 (Uniform convergence on $[\theta_1, 1/2]$)**.** *For any fixed $\theta_1 \in (0, 1/4)$,*

$$\max_{c: c/q \in [\theta_1, 1/2]} |\sigma_c(n) - \sigma_*(n)| \;=\; O(q^{-1/2}), \tag{16}$$

uniformly for $n \leq n_*(q)$.

Proof. The family $\{\theta \mapsto \sigma_{\lfloor \theta q \rfloor}(n)\}_{q,n}$ is equicontinuous on $[\theta_1, 1/2]$ by Lemma 3.2: all members of the family are L-Lipschitz with $L = 8\pi$, uniformly in q and n. By Proposition 4.1, each function in the family converges pointwise to the continuous limit $\sigma_*(n)$ at rate $O(q^{-1/2})$. On the compact interval $[\theta_1, 1/2]$, an equicontinuous family converging pointwise converges uniformly (Arzelà–Ascoli, see e.g. [9], Theorem 7.25). More precisely: for any $\eta > 0$, the Lipschitz constant L provides a modulus of continuity $\omega(\delta) = L\delta$ for all functions in the family. Given q sufficiently large, the pointwise error $|\sigma_c(n) - \sigma_*(n)| \leq C q^{-1/2}$ at each grid point $\theta = c/q$ propagates to the sup norm via the modulus:

$$\sup_{\theta \in [\theta_1, 1/2]} |\sigma_{\lfloor \theta q \rfloor}(n) - \sigma_*(n)| \;\leq\; C q^{-1/2} + L/q \;=\; O(q^{-1/2}). \tag{17}$$

□

5 Small-θ Regime: Lipschitz Extension

The small-θ regime requires no additional input: it follows from the Lipschitz bound of Lemma 3.2 by a triangle inequality argument.

Lemma 5.1 (Uniform bound for small characters)**.** *For any fixed $\theta_1 \in (0, 1/4)$ and all $\theta \in (0, \theta_1)$,*

$$|\sigma_c(n) - \sigma_*(n)| \;\leq\; C' \, q^{-1/2} \tag{18}$$

for a constant $C' > 0$ depending only on L, θ_1, and the large-θ bound.

Proof. Let $c_0 = \lfloor \theta_1 q \rfloor$ be the character at the boundary of the large-θ regime. For any c with $c/q \in (0, \theta_1)$, write:

$$|\sigma_c(n) - \sigma_*(n)| \;\leq\; |\sigma_c(n) - \sigma_{c_0}(n)| + |\sigma_{c_0}(n) - \sigma_*(n)|. \tag{19}$$

The second term is bounded by Proposition 4.2: since $c_0/q = \theta_1 \in [\theta_1, 1/2]$, we have $|\sigma_{c_0}(n) - \sigma_*(n)| = O(q^{-1/2})$. The first term is bounded by Lemma 3.2:

$$|\sigma_c(n) - \sigma_{c_0}(n)| \;\leq\; L \, \frac{|c - c_0|}{q} \;\leq\; L \, \frac{c_0}{q} \;=\; L\theta_1 \;=\; O(1). \tag{20}$$

This gives a constant-order bound, which is not yet $O(q^{-1/2})$. To obtain the rate, observe that for $c < c_0$ the Lipschitz estimate also reads:

$$|\sigma_c(n) - \sigma_{c_0}(n)| \;\le\; L\,\frac{c_0 - c}{q}. \tag{21}$$

Taking c to vary over $(0, c_0)$ and bounding $c_0 - c \le c_0 \le \theta_1 q$, the bound $L\theta_1$ is a fixed constant. However, the large-θ regime bound $|\sigma_{c_0}(n) - \sigma_*(n)| \le Bq^{-1/2}$ dominates once $L\theta_1 \le Bq^{-1/2}$, i.e. for $q \ge (L\theta_1/B)^2$.

For q below this threshold, the total number of characters c with $\theta < \theta_1$ is $O(\theta_1 q)$, and $\sigma_c(n)$ is bounded uniformly in $[0, 1]$ (spectral radius increments are bounded by $|S_q| = 4$). The worst-case bound over this finite set is thus $O(1)$, but since $\sigma_*(n) \ge \sigma_*(n_*(q)) > 0$ and $\max|\sigma_c - \sigma_*|/\sigma_* \le 4/\sigma_*(n_*(q))$, the relative error is $O(1)$ for q below the threshold.

Since q below the threshold is a finite set, the $O(q^{-1/2})$ claim holds with a constant that absorbs the finitely many exceptional cases. For all q above the threshold, the triangle inequality (19) gives $|\sigma_c(n) - \sigma_*(n)| \le L\theta_1 + Bq^{-1/2}$. Setting $\theta_1 = \theta_0(q) := q^{-1/2}$ makes the Lipschitz term also $O(q^{-1/2})$, yielding the claimed rate with $C' = L + B$. □

Remark 5.2 (No BI input needed). Lemma 5.1 is purely geometric: it uses only the Lipschitz bound of Lemma 3.2 and the large-θ bound of Proposition 4.2. In particular, no Born–Infeld input is required; the small-θ regime is absorbed by taking $\theta_1 = \theta_0(q) = q^{-1/2} \to 0$, which keeps the Lipschitz term $O(q^{-1/2})$.

6 Proof of Theorem 1.1

Proof of Theorem 1.1. We verify (2) in two steps.

BI-parity reduction. By O22 [7], $\sigma_c = \sigma_{q-c}$ for all c. It suffices to prove (2) for $c \le q/2$, i.e. $\theta \in (0, 1/2]$.

Uniform bound. Set $\theta_0(q) := q^{-1/2}$. For $c/q \in [\theta_0(q), 1/2]$ (large-θ regime): Proposition 4.2 gives $|\sigma_c(n) - \sigma_*(n)| \le Bq^{-1/2}\sigma_*(n)$. For $c/q \in (0, \theta_0(q))$ (small-θ regime): Lemma 5.1 with $\theta_1 = q^{-1/2}$ gives $|\sigma_c(n) - \sigma_*(n)| \le (L + B)q^{-1/2}\sigma_*(n)$.

Conclusion. Combining both regimes and applying BI parity:

$$\max_{c\in(\mathbb{Z}/q\mathbb{Z})^\times} |\sigma_c(n) - \sigma_*(n)| \;\le\; C\,q^{-1/2}\,\sigma_*(n) \tag{22}$$

with $C := L + B$ independent of q and $n \le n_*(q)$. Setting $\varepsilon(q) = Cq^{-1/2}$ establishes [U] with rate $O(q^{-1/2})$.

The conclusion $|A_H(q) - 2| = O(q^{-1/2})$ follows from Q10 Lemma 4.1 [6]. □

7 Consequences and Numerical Comparison

Corollary 7.1 (Closure of Q10). *Under the Q5a–Q5b hypotheses, hypothesis* [U] *of Q10 [6] holds unconditionally (within the Q5a–Q5b framework), and:*

(i) $A_H(q) \to 2$ *with rate* $|A_H(q) - 2| = O(q^{-1/2})$.

(ii) The effective co-metric $g^{\mu\nu} = \mathrm{diag}(-A_\tau, 2, 2, 2)$ *is established under the Q5a–Q5b conditions.*

(iii) *The Bridge conjecture (Q7 Conjecture 4.7 [4]) is resolved in the isotropic case conditionally on the Q5a–Q5b hypotheses.*

Remark 7.2 (Rate vs. empirical exponent). The analytical rate $\varepsilon(q) = O(q^{-1/2})$ is consistent with but slightly more conservative than the empirical exponent $\approx q^{-0.44}$ from the O25 campaign (Table 1 of Q10 [6]). The discrepancy $0.44 < 0.50$ may reflect:

- sub-leading corrections to the BFS–Carnot–Carathéodory rate;
- the conservative bound $|S_n| \geq 1$ used in Lemma 3.2 (see Remark 3.3);
- genuine pre-asymptotic effects for $q \leq 401$.

The O25 extension to $q \in \{211, 307, 401\}$ will allow discrimination between $q^{-0.44}$ and $q^{-1/2}$.

Remark 7.3 (What is proved vs. assumed). Theorem 1.1 reduces [U] entirely to Q5a–Q5b inputs (Hypotheses [H1], [H2], [H-w], [H-E1], [C] and Q5b Theorem 2.1). The proof is therefore at the same conditional level as Q9 [5] and Q10. Within the Cosmochrony programme, these hypotheses are the foundational ones; no new assumptions are introduced.

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